

Editor's email

14-Aug-2025

MS: MBE-25-0700

Title: Seasonal influenza viruses show distinct adaptive dynamics during growth in chicken eggs

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We look forward to receiving a revised version of the manuscript for further consideration.

Sincerely,

Board of Editors
Molecular Biology and Evolution

Reviews

Associate Editor

Editors' comments to the author:

Two reviewers have seen the paper and both agree this is generally an excellent paper. Reviewer 2 does have some concerns about validity of some analyses and quality of presentation but I am sure the authors can address these.

Reviewers' comments:

Reviewer: 1

Comments to the Author

In this study, Kistler and Bedford performed a systemic analysis of egg-adaptive mutations in four seasonal influenza viruses (A/H3N2, A/H1N1pdm, B/Vic and B/Yam). The authors showed that non-synonymous mutations during egg-adaptation were enriched at the receptor binding site of HA compared to other segments, suggesting the critical function of egg-adaptive mutations for receptor specificity. Importantly, egg-adaptive mutations of A/H3N2 HA, but not A/H1N1pdm, B/Vic and B/Yam, are dependent on natural variants. Therefore, egg-adaptive trajectory changes as the virus evolves. The three explanations of the time-dependence of egg-adaptive mutations are well thought-out. Overall, the manuscript addresses an important question with high public health relevance, as egg-derived seasonal vaccines still account for more than half of the vaccine production. I only have minor comments.

Minor comments:

1. Page 11: "prolines limit rotation around the protein background". Should that be "protein backbone" instead?

Yes- thank you! The edit has been made.

2. Page 13: "Both of the 190 mutations act". Should that be "190-helix mutations" instead?

Good suggestion. We have changed this phrase of the text to "Both of the mutations in the 190-helix..."

3. Page 14 line 3, does "275" refer to the number of strains with F195?

Ah, yes 275 refers to the number of human H3N2 strains with 195F that were passaged in eggs. We have attempted to clarify this by changing the sentence to: "After this point, 186N mutations and reversions to 195Y both become frequent in egg-passaged strains, occurring in 94/275 and 29/275 of the egg-passaged strains derived from unpassaged strains with 195F, respectively"

4. Figure 5 has 5 panels (A-E), but its legend only describes two panels (A and B).

Oops. Thank you for pointing this out! We have fixed the legend.

Reviewer: 2

Comments to the Author

This study investigates the fitness landscapes of four seasonal influenza viruses by analysing egg-adapted virus strains, which have been generated as part of vaccine production. For a significant proportion of the egg-adapted viruses, sequences are available for the original viruses (unpassaged). Additionally, sequence data also exists for multiple, independent, egg-passaged viruses from the same strain. Consequently, this study provides a unique opportunity to understand emergence of adaptive mutations and how this affected is affected by evolutionary constraints across the different seasonal viruses.

Most adaptive mutations were found to occur in the receptor binding domain of HA1 in the haemagglutinin (which is likely driven by different sialic acid receptors being present in chicken eggs compared to humans). Background genotype, which changes over time due to continual antigenic evolution, play a critical role in appearance and persistence of mutations. As H3N2 subtype is characterised by faster rate of antigenic evolution, these viruses appear to be most influenced by background genotype, followed by H1N1pdm viruses. In contrast, the same pattern is not observed in influenza B viruses. Additionally, a higher frequency of multiple co-occurring mutations was observed in H3N2 viruses (~50%) compared to the other seasonal viruses. Lastly, by looking at sequences of multiple independently passaged viruses derived from the same strain, the authors reveal that several evolutionary pathways are available to the virus, suggesting the emergence of adaptive mutations is not purely deterministic but also influenced by stochastic effects (and further suggests most adaptive mutations do not have large fitness effects).

Overall, I thought this study's approach is innovative and reveals new and important insights into the molecular evolutionary dynamics of seasonal influenza viruses. However, I have some questions about the methodology and interpretation of results, which I have outlined below.

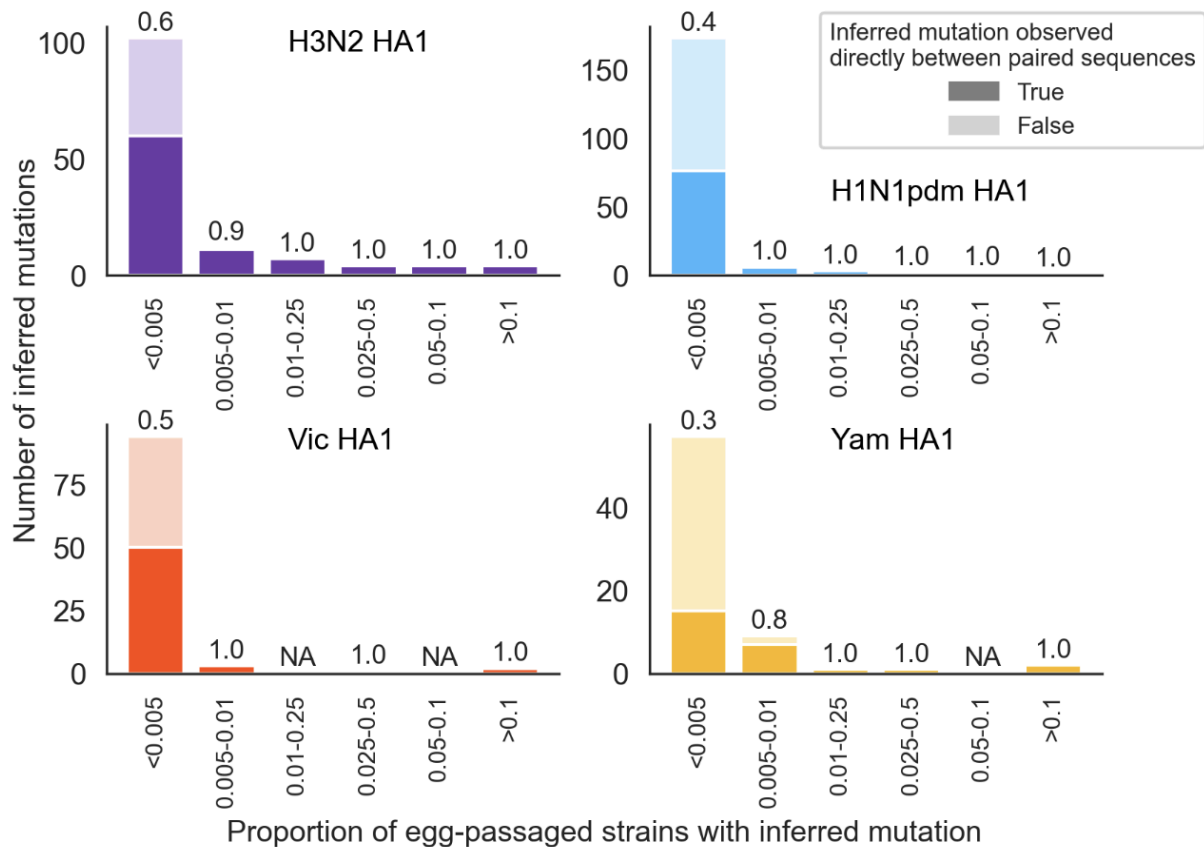
1) The methodology to infer egg-adaptive mutations is clearly described in Figure 1F. I can understand how mutations can be confidently inferred for egg-adapted viruses associated with short branches (i.e. when egg-passaged viruses cluster with the original or closely related unpassaged virus). However, for longer branches, it is possible that some mutations in the passaged strains might have originally arose in the unpassaged viruses, raising the possibility that some egg-adaptive mutations might include false positives.

Did the authors consider the impact of long branches on inferring egg-adaptive mutations? If not, it might be worthwhile to compare the extent of overlap of mutations identified from “pairs” versus “non-pairs”. If false positives have minimal impact, there should be a large overlap between “pair” and “non-pair” mutations. Alternatively, the authors may want to consider applying a branch-length threshold to exclude inferences with high uncertainty.

It is a good point that some mutations may actually have occurred in unsampled (or not included in tree) unpassaged strains. In this case, mutations that are not fixed in human influenza and occur only very rarely may be falsely attributed as occurring during egg-passaging. Because of this, these “false positive” mutations should occur in only one or a few egg-passaged strains. Thus, because our analyses of egg-adaptation only consider mutations that occur in a large number of strains, we actually have excluded these potential “false positives” from our analyses.

Here, we’ve done something similar to the reviewers’ suggested “pair” vs “non-pair” analysis. Using just the egg-passaged strains that were sequenced prior to egg-passaging, we found all “known” egg-passaging mutations via a direct comparison of the HA1 sequence of an egg-passaged strain and its unpassaged pair. We then compare these mutations to the egg-passaging mutations we inferred from the tree (for ALL egg-passaged strains). We see here that the inferred and “known” mutations agree for all common mutations (in fact, for all mutations that occur in 1% or more of egg-passaged strains). Thus, the inferred mutations that were not directly observed between pairs are rarer mutations, occurring in just one or a handful of egg-passaged strains. This suggests that these mutations either actually arose in a human virus, or that they are just random (unselected) mutations that arose during egg-passaging. Either way, we do not consider any of these mutations to be potentially adaptive and none of our analyses of adaptation include them.

We have included Review Figure 1 as Supplemental Figure 1 in the text and added text to the first section of the Results describing it.



Review figure 1: Inferred egg-passaging mutations largely overlap with mutations directly observed between egg-passaged/unpassaged pairs. All HA1 residues where we inferred a mutation during egg-passaging via our tree-based method were binned by the proportion of egg-passaged strains they occurred in. Within each bin, the total number of inferred mutations that are also observed via direct comparison of egg-passaged/unpassaged sequence pairs is plotted in dark shading, while inferred mutations that are not directly observed between pairs is plotted in light shading. For each bin, the proportion of inferred mutations that are observed directly between paired sequences is written above the bar.

2) Mutations arising due to relaxation of pressures – while these mutations are briefly mentioned in the introduction, it does not appear these mutations were explicitly included in the analysis. Is that correct? Reversion mutations can also provide useful insights about the fitness landscape. For example, I was wondering if mutations identified in NP could represent reversions, especially if they overlap with known immune epitopes. It would be helpful to clarify if reversion mutations were considered in this study, and whether any analysis was performed to differentiate adaptive versus reversion events.

As the reviewer mentions, we suggest that relaxation of certain selective pressures present in humans might result in reversion mutations during egg-passaging. In our analysis, we look at *all* mutations that occur during egg-passaging. This includes reversion mutations though we do not explicitly categorize mutations by reversion or non-reversion. The reason we do not draw attention to this category of mutations is because we actually see multiple possible hypotheses

as to why reversion mutations would occur, and thus, this category doesn't seem to offer additional explanatory power.

For instance, several of H3N2's adaptive HA mutations (156Q, 160K, 186S, 195Y, and 225G/N) could be considered reversions. It is unclear, however, whether these reversions represent a relaxation of some selection or that the genotype of the human virus is actively being selected against in eggs, or even a combination of both. The 160K mutation is a good example. Human influenza had 160K until around 2013 at which point 160T fixes and circulates until around 2020 when it is replaced by 160I. During the period where human H3N2 has 160T, the majority egg-passaged strains get 160K or 160I mutations. Threonine at residue 160T adds a glycosylation motif and the presence of this glycan has been shown to alter receptor-binding specificity towards longer sialic acids ([Thompson et al, 2024](#); [Bolton et al, 2022](#)). The inability of H3N2 virus with 160T to bind short sialic acids, such as alpha-2,3 linked sialic acids that are found in eggs, is the likely cause of its poor growth in eggs ([Zost et al, 2017](#)) and the strong selection for mutations that ablate the glycosylation motif. Here, we observe a reversion mutation where the underlying source of selection in eggs is that the human influenza virus fixed a mutation that is maladaptive in eggs.

The other H3N2 HA reversions do not add or remove glycans, but these mutations show similar patterns of occurring on particular human influenza genotypes. As more example: egg-adaptive mutations 156Q and 156R happen when virus with 156H is passaged in eggs. The human virus has had many fixations at this position: 156K->Q->H->S. Thus, the egg-adaptive mutations at this position could be explained by 156H being maladaptive in eggs (and so there is selection for egg mutations at this position when the human virus has 156H). Equally, this could be explained if 156H is actually not particularly favorable for any HA, but that it confers such strong immune escape that it is selected in humans anyways; however, when the virus is passaged in eggs, 156 reverts due to a relaxation of that immune selection.

The reviewer also mentions NP mutations. Within the data we use for this manuscript, we find that none of the H1N1pdm NP mutations are reversions and one of the two H3N2 mutations (384R) is a reversion. The H3N2 tree we use in this study only dates back to 1995 and H1N1pdm was only introduced in 2009, yet the NP segment of both of these viruses actually originates from the 1918 H1N1 pandemic virus. Therefore, the present-day influenza NP proteins should be heavily human-adapted compared to the 1918 introduction from birds. Because of this, we also compare the egg-adaptive mutations to the sequence of the 1918 H1N1 NP. However, this actually does not change the results (H3N2 384R is the only reversion).

We added the following text to the Results section (context shown in italics):

Fitting with a potential role in host adaptation, all 5 of these residues are found amongst the sites that Kukol and Hughes identify as varying most between hosts ²⁶. Though it may be expected that egg-adaptation in NP would result in reversion mutations that revert the protein back from human-adapted to avian adapted. The NP segment of both H3N2 and H1N1pdm was introduced from avians to humans in the 1918 H1N1

pandemic and was subsequently transferred multiple times through different subtypes and different species via reassortment. We, therefore, compare the egg-adaptive NP mutations to a 1918 H1N1 NP sequence (Genbank AY744935) and find that only one (H3N2 G384R) of the five mutations we identify is a reversion.

The NP protein is essential for transporting viral RNA into the nucleus for replication²⁷, a process that relies on interactions with importins and other host cell machinery that differ between humans and birds²⁸.

3) Page 7, line 32, “to” missing after “pressure”

Fixed. Thank you!

4) Page 10, line 28 – “at eggs”, should read “in eggs”

Fixed. Thanks!

5) I found Figure 4 hard to parse and took several readings in conjunction with Figure 3 to fully understand the results in the main text, which goes into some detail explaining the results. To improve clarity, would it be possible to present a subset the trees in Figure 4 to highlight key results discussed in the main text. It would be helpful to also include a plot of the frequencies of the background and egg-adapted mutations over time (similar to Figure 3) for each tree.

We have separated Figure 4 into two figures (now Figure 4 and 5), which we hope will help with the readability and interpretability of the figure. Figure 4 now focuses on egg-adaptive mutations that depend on the genotype of the human virus at the same codon and Figure 5 shows egg-adaptive mutations that show epistatic dependence on the genotype of the human virus at a different residue. As the reviewer suggested, we added frequency plots to show the frequency of human genotypes over time along with the occurrence of egg-adaptive mutations.

For the epistasis results, I would suggest presenting the subplots of the location of mutations in parallel with trees/frequency over time plots.

In the new Figure 5, we have aligned the panels so that the phylogeny, frequency plot, and structure of each epistatic interaction are in a single column.

Adjusting the colour palette as orange and red in the trees are hard to distinguish.

We have adjusted the orange to a yellow, which is more distinguishable from the red.

6) The frequency of adaptive substitutions across the four viruses mirroring the rates of antigenic evolution doesn't seem particularly striking given that background genotype influences the likelihood of adaptation and that background genotype diversity itself increases with antigenic evolution. Therefore, I am not convinced on these analyses alone that H3N2 viruses are more tolerant to mutations compared to other subtypes. To more directly assess whether H3 viruses are genuinely more mutationally flexible/tolerant, additional analyses on H1 and H3 viruses in other host species will be necessary, e.g. swine or avian. I am not suggesting such analyses are necessary for this study, but I recommend that the limitations of this interpretation be discussed more explicitly.

The reviewer is suggesting that the correlation between rate of antigenic evolution and the number of egg-adaptive mutations observed is due to the former causing the latter. More explicitly, the hypothesis is that faster antigenic evolution results in a faster changing background and, thus, in more changes in which mutations are egg-adaptive. We think that this is definitely a contributing factor but we do not think this explains the whole story.

The primary reason for this is because we observe that H3N2 strains with identical sequences can get up to 8 different egg-adaptive mutations, while H1N1pdm strains with identical sequences can get up to 4, and identical sequences of influenza B viruses can get up to 3 (Figure 7B). This means that a higher number of egg-adaptive mutations are available to H3N2, then H1N1pdm, then influenza B viruses, in a background-independent (and thus antigenic evolution independent) manner. Supplementary Figure 9A offers further support of the background-independence of this relationship: we observe that more mutations are consistently available to these viruses within 3-year windows of time during which the background should be largely the same.

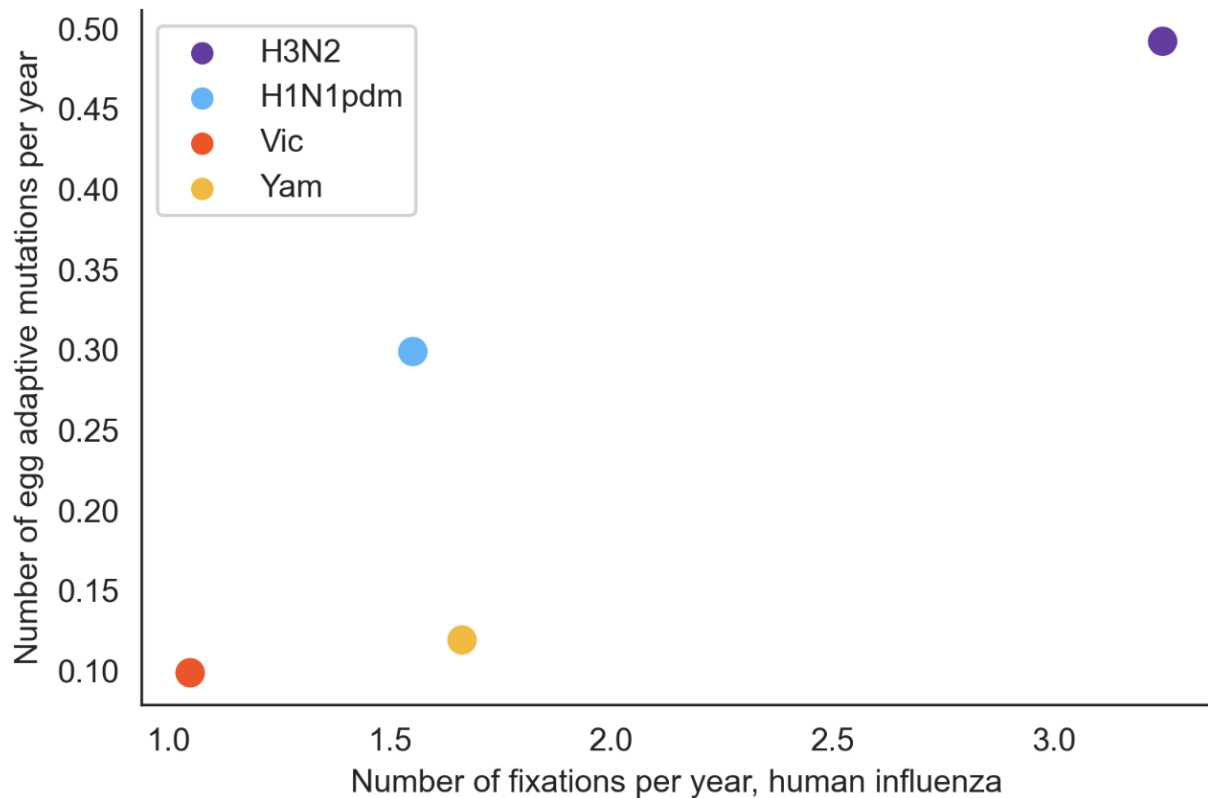
The reviewer's comment gave us the idea to do another analysis related to this question: here, we compare the number of HA1 fixations per year to the number of egg-adaptation mutations per year (Review Figure 2). In this figure, we see that H3N2 has the highest rate of fixations in the human virus, and the highest number of egg-adaptive mutations (even when accounting for the amount of time these mutations are observed over). However, H1N1pdm and Yam both have similar rates of fixations in the human virus, while H1N1pdm gets egg-adaptive mutations at more sites than Yam. This offers another small piece of evidence against the hypothesis that the correlation between rate of antigenic evolution and number of egg-adaptive mutations is solely driven by the former causing the latter, though we don't think this analysis is super strong.

The reviewer also suggested looking at the evolution of avian H3 and H1 for additional clarification on whether H3 is more tolerant of mutations than H1. We think this is a great idea, however, we don't see evidence of H1 or H3 evolving adaptively in birds (see Review Figure 3 and an explanation under the reviewer's next comment).

We have added Review Figure 2 as Supplemental Figure 12 in the manuscript. We have also added a sentence to the Discussion to draw more attention to the results presented in Figure 7B, and their implication that faster antigenic evolution alone cannot explain higher numbers of egg-adaptive mutations (a contextual sentence are in italics):

The ongoing antigenic evolution in humans incidentally makes the virus more or less fit when it is subsequently passaged in chicken eggs, and also can alter available pathways towards egg-adaptation (Figures 3-7). Importantly, faster antigenic evolution (which alters the background), alone, cannot fully explain the higher number of egg-adaptive mutations observed among H3N2, then H1N1pdm, then influenza B viruses because we observe this trend amongst viral strains with identical sequences (Figure 7B) and because we observe that A/H1N1pdm viruses get more egg adaptive

mutations than B/Vic viruses despite both accruing a similar number of fixations per year in humans (Supplemental Figure 12).



Review Figure 2: Correlation between rate of human influenza HA1 background fixations and observed number of egg-adaptive mutations. The number of fixations observed throughout HA1 in human viruses during the same time period as the egg-adaptation analysis was calculated from a tree containing only samples of human virus. The total number of HA1 sites where egg-adaptive mutations are observed is normalized by the total amount of time, to account for the fact that H1N1pdm only entered humans in 2009 and Vic went extinct in 2020.

7) On a related note, as noted egg-adapted mutations are not usually found at same sites in avian H1 and H3 viruses compared to human viruses, do you think you would find a similar pattern of higher mutations in H3 vs H1 in avian viruses or alternatively would you expect find patterns to be more consistent with findings for influenza B viruses? A brief discussion on this point would add depth to the evolutionary implications of these findings.

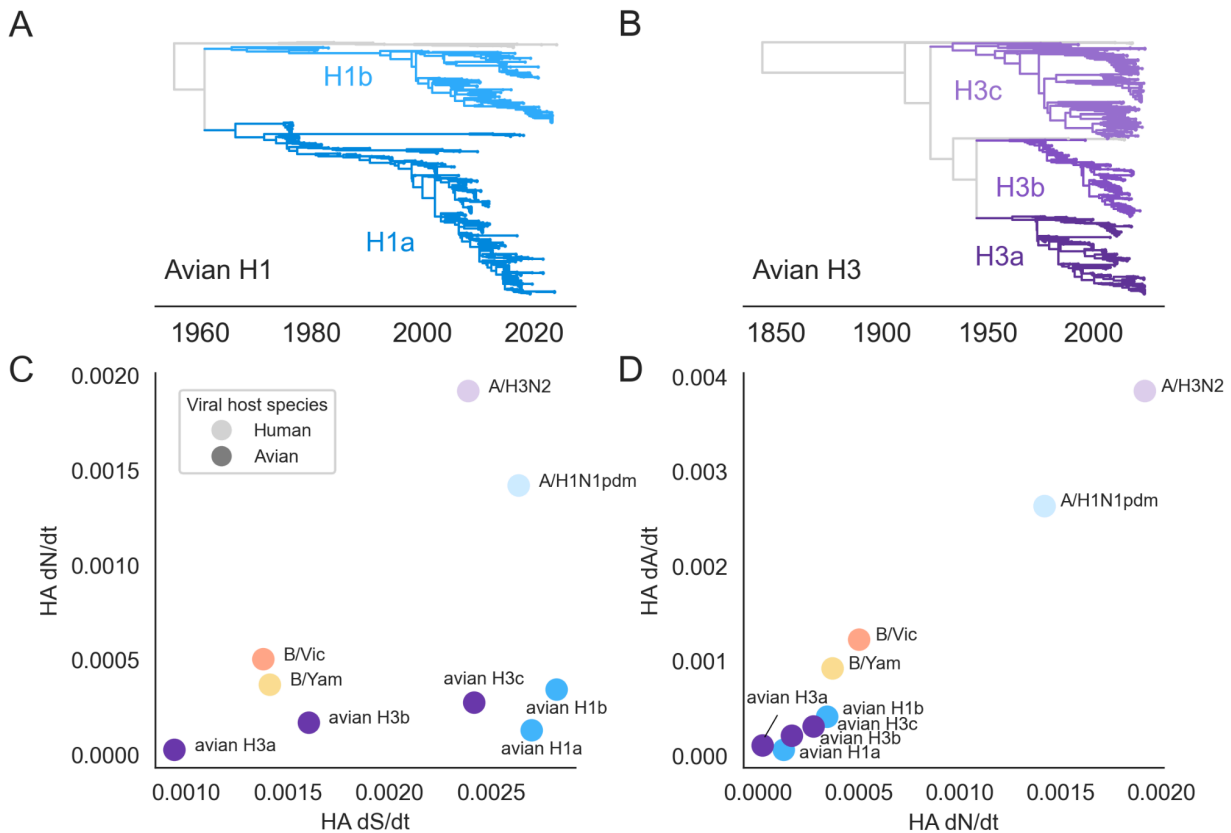
To address whether we observe similar patterns of evolution between avian H1 and H3 viruses as we do with human H1 and H3 viruses, we calculated rates of nonsynonymous and synonymous mutation, as well as rates of adaptive evolution for avian H1 and H3 and compare these to the rates we observe in human influenza viruses. Both avian H1 and H3 consist of multiple, co-circulating lineages (see: <https://nextstrain.org/groups/blab/avian/H1> and <https://nextstrain.org/groups/blab/avian/H3>), so we calculate the rates for these lineages separately. We do not observe convincingly different rates of adaptive mutations in avian H3

versus H1, and in fact, the rates for the avian viruses are much lower than those of the human viruses.

We think the low rates of nonsynonymous mutation and adaptive evolution that we observe in avian versus human viruses supports a lack of ongoing antigenic evolution in these avian viruses. In the absence of adaptive evolution, it is impossible to say whether avian H3 has access to more potentially adaptive mutations than avian H1. It is outside of the scope of this paper to pin down exactly why avian H1 and H3 do not evolve antigenically, but we hypothesize that it could be some combination of: 1) shorter lifespan of avian versus human hosts, meaning that a greater proportion of the host population has not had a prior infection, 2) wide diversity of hosts (avian influenza viruses infect many families, and sometimes multiple orders, of birds) that may have slightly different predominant sialic acid types, 3) co-circulation of 16 HAs and 9 NAs with prolific reassortment, potentially allowing the virus to readily re-infect hosts via reassortment-mediated diversity rather than via antigenic evolution.

We added Review Figure 3 to the paper as Supplemental Figure 10, and also added the following text to the Discussion (contextual sentences in italics):

Interestingly, this same ordering of subtypes also describes their relative rates of antigenic drift^{34,35}, offering the possible hypothesis that this H3 hemagglutinin has more mutational flexibility in the receptor-binding pocket, which allows it to more readily adapt to $\alpha 2,3$ receptors and also to more readily access antigenic variation that does not compromise receptor-binding. If this hypothesis is true, we should also observe faster adaptation of H3 compared to H1 in other contexts, such as in other species. However, we do not find evidence of ongoing adaptation within avian viruses (Supplemental Figure 10) and therefore cannot make this comparison between avian HAs. From our results, we cannot distinguish between the potential explanations of the different egg-adaptation observed between influenza subtypes.



Review Figure 3: Avian influenza A viruses have slower rates of nonsynonymous mutation and adaptive evolution than their human counterparts. A) Phylogeny of 899 avian H1 sequences, showing our designation of lineages H1a and H1b. B) Phylogeny of 1284 avian H3 sequences, showing our designation of lineages H3a, H3b, and H3c. C) Comparison of rates of synonymous and nonsynonymous mutation in hemagglutinin (HA) of four human influenza viruses (lighter shading) and five lineages of avian influenza A (darker shading). D) Comparison of the rate of nonsynonymous mutation and rate of adaptation evolution in HA of the same human and avian influenza viruses. Rates of synonymous and nonsynonymous mutation (per nucleotide per year) are calculated relative to the inferred ancestral sequence of that viral species, or lineage. Rates of adaptation across HA (in units of adaptive mutations per codon per year) are calculated according to the method in Kistler and Bedford 2023 (<https://www.sciencedirect.com/science/article/pii/S1931312823003803>).

8) I was also curious about whether variation in background genotype across the four viruses has implications for the ability to detect egg-adaptive mutations? It seems that an alternative explanation for finding lower egg-adaptive mutations or epistasis in influenza B viruses could be because the limited background genotype diversity limits the power to detect such mutations.

We agree that lower background variation affects the occurrence of egg-adaptive mutations, as the human virus inadvertently explores genotypes that are more or less fit in eggs (this is discussed in the response to Reviewer's comment 6, above). However, we do not think lower genetic diversity amongst human influenza strains will affect the ability to *detect* egg adaptive mutations. If anything, it should make this analysis easier and more robust. For example,

consider the extreme example of a uniform genotype of human influenza viruses. In this case, any differences observed in the sequence of this human virus and the egg-passaged strains will have occurred in eggs. The more diverse the human virus becomes, the harder it is to disentangle unsampled human influenza diversity from mutations that occurred during egg-passaging (this is discussed under Reviewer's first comment, above).